

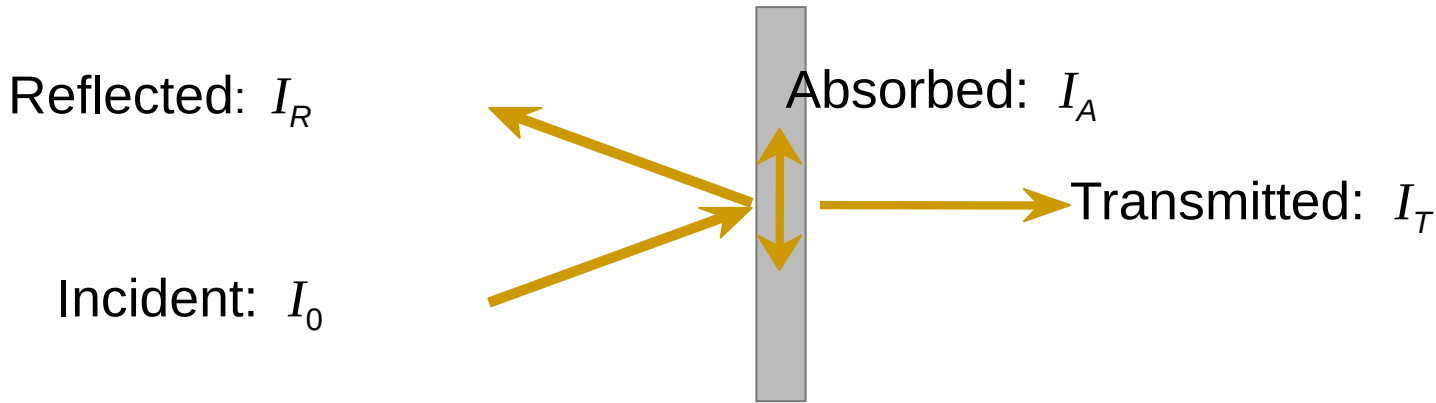
Karakterisasi Material:

Spektroskopi UV-Vis dan PL

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Interaksi Cahaya-Material



$$I_0 = I_T + I_A + I_R$$

$$T + A + R = 1$$

$$T = I_T / I_0$$

$$A = I_A / I_0$$

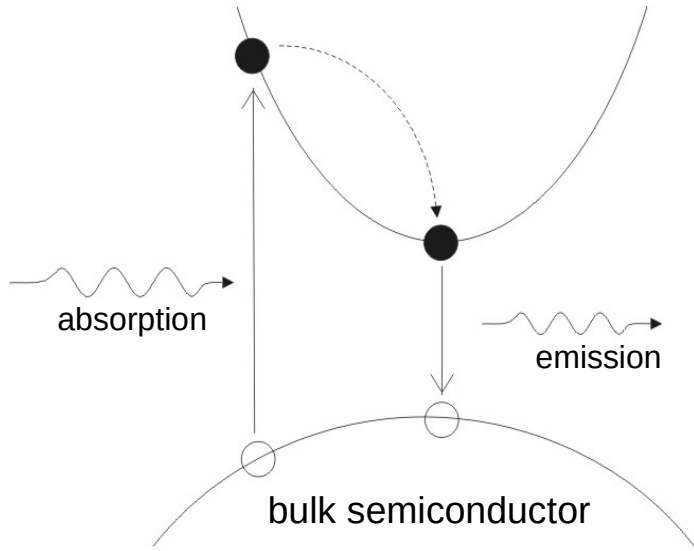
$$R = I_R / I_0$$

Optical Characterization methods

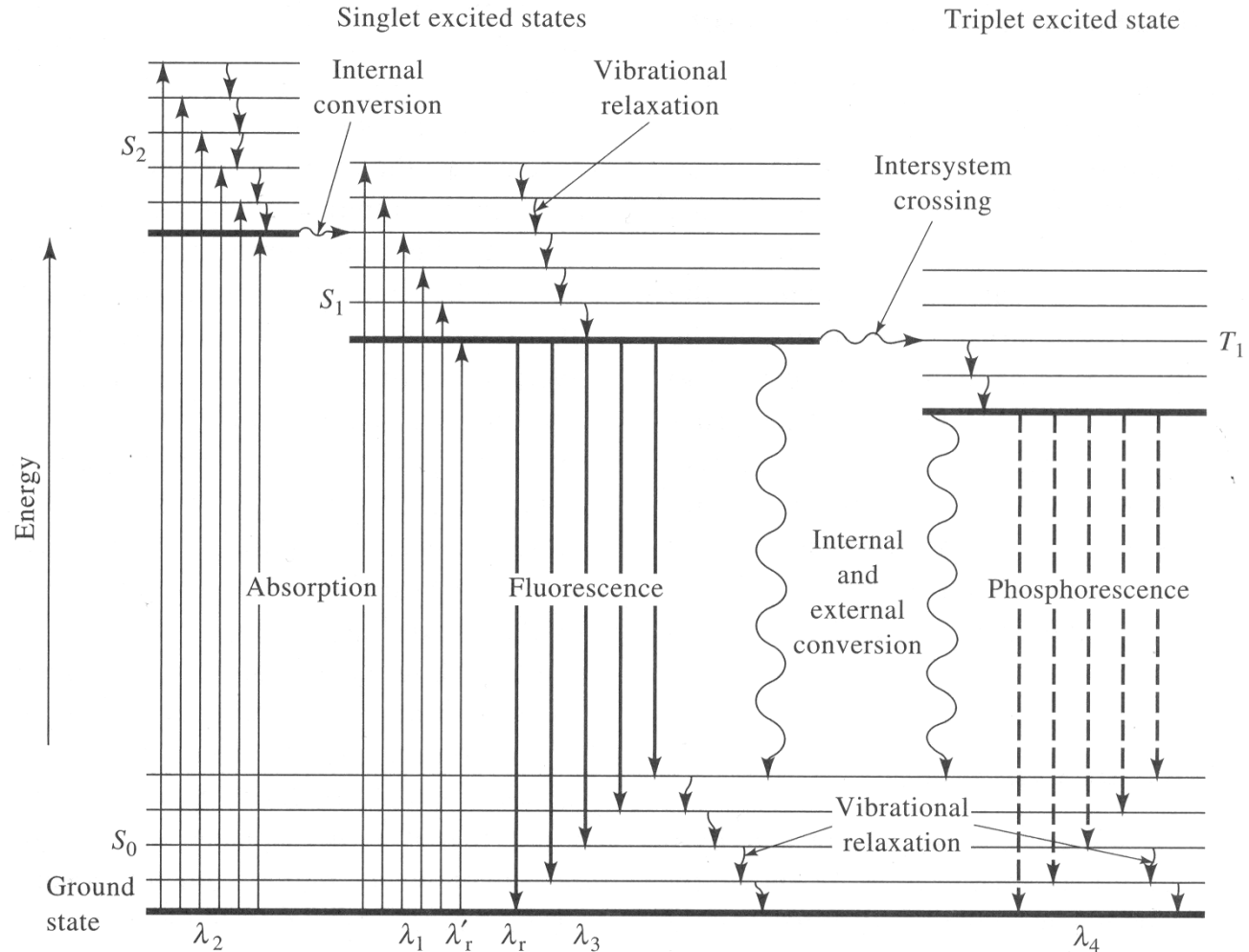
Technique	Synopsis	Solid State Physics
Absorption spectrometry	Scan λ_{in} , measure intensity of transmission.	Optical absorption, band gap, energy level spacing
Photoluminescence (PL)	Fixed λ_{in} (laser), scan λ_{out}	Optical recombination transitions.
Photoluminescence Excitation (PLE)	Fix λ_{out} , scan λ_{in} (tunable laser or monochromator)	Sensitive to transitions that “pump” optical emissions.
Raman scattering	Laser in, scan λ_{out} very close to λ_{in}	Stokes/anti-Stokes peaks provide information about phonon energies.

- Transmission: “untouched” photons
- PL includes fluorescence and phosphorescence.
- Emission wavelength usually longer than excitation wavelength (Stokes shift)

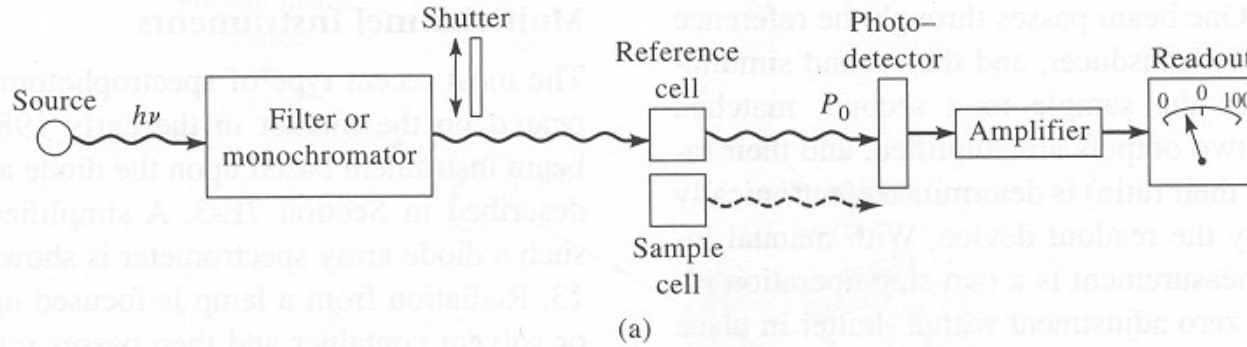
Energy levels



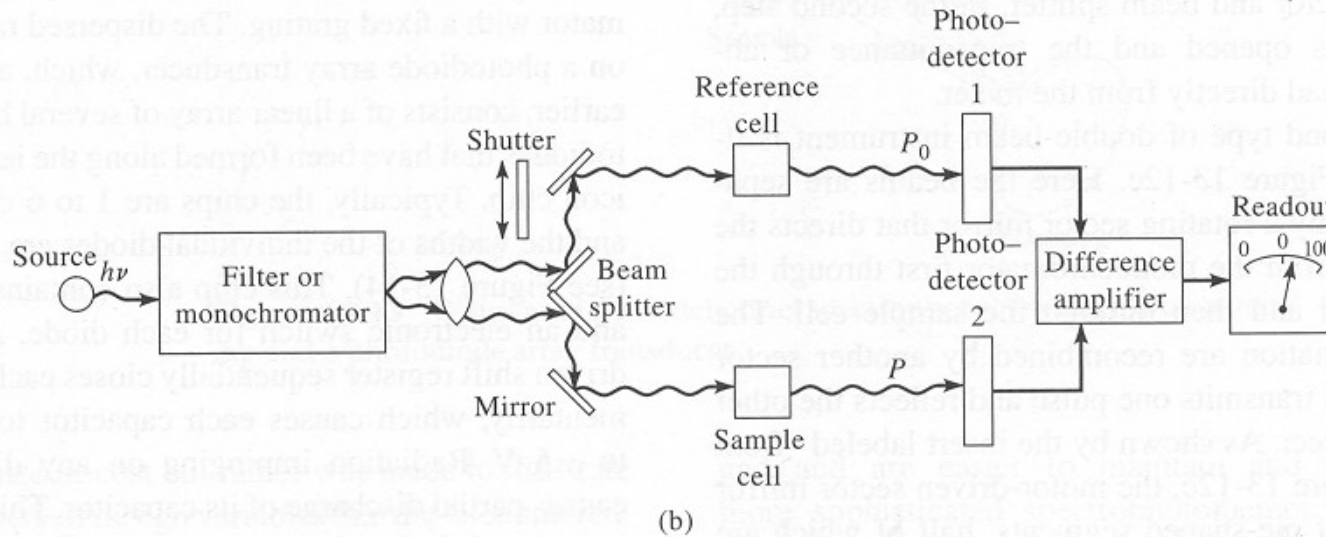
PL semikonduktor curah (bulk) biasanya memiliki puncak pada celah pita, sedangkan absorpsi dan PLE dapat menentukan rapat keadaan.



Absorption Spectrometry



Untuk instrumen single beam, data diperoleh dua kali, sekali dengan sel referensi, sekali dengan sel sampel di tempatnya. Jatah sinyal diambil untuk memberikan absorbansi.



Double beam menyetel nol dengan shutter tertutup; ketika shutter dibuka absorbansi dibaca langsung dari perbedaan amplifier.

Kedua setup memiliki (1) filter atau monokromator untuk pemilihan panjang gelombang, (2) transduser dan (3) perangkat pembacaan untuk pengumpulan data.

Contoh spektrum

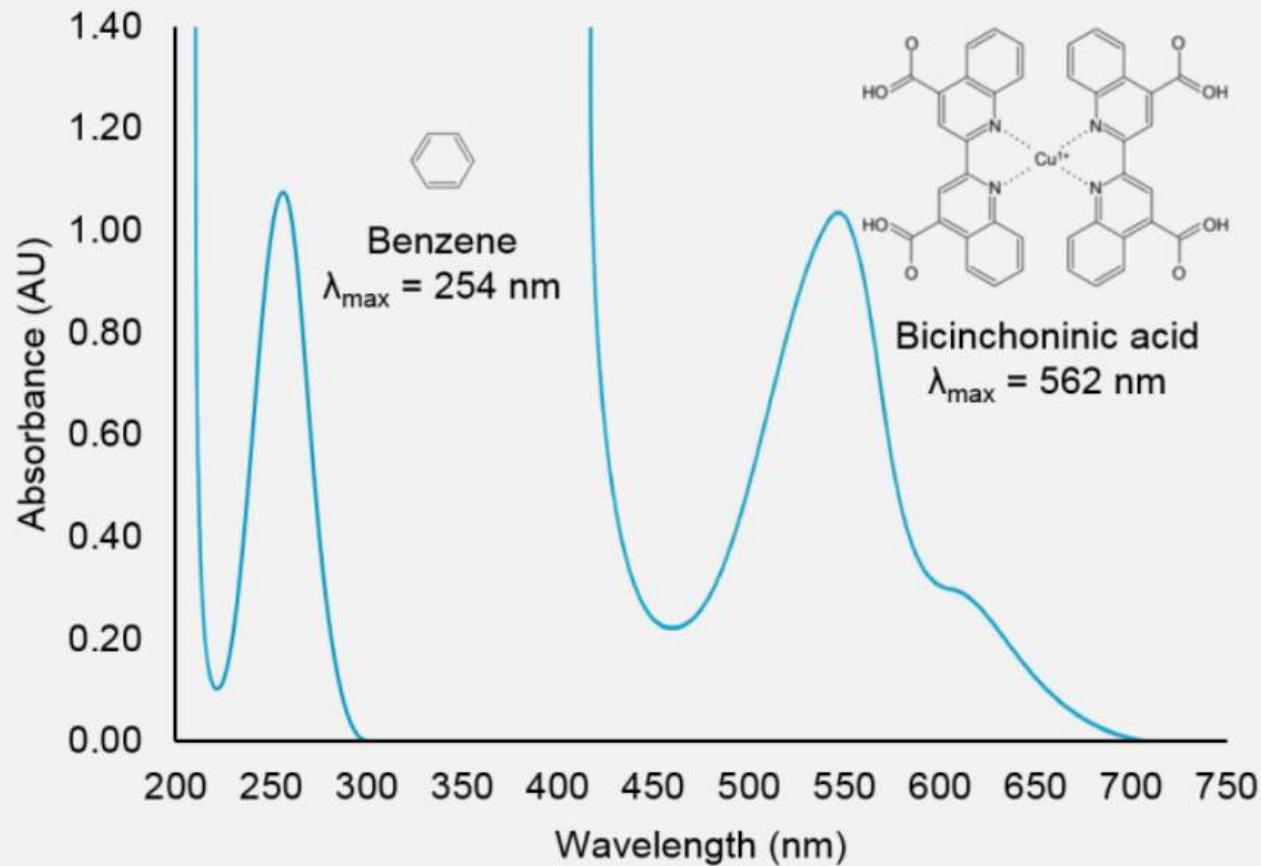
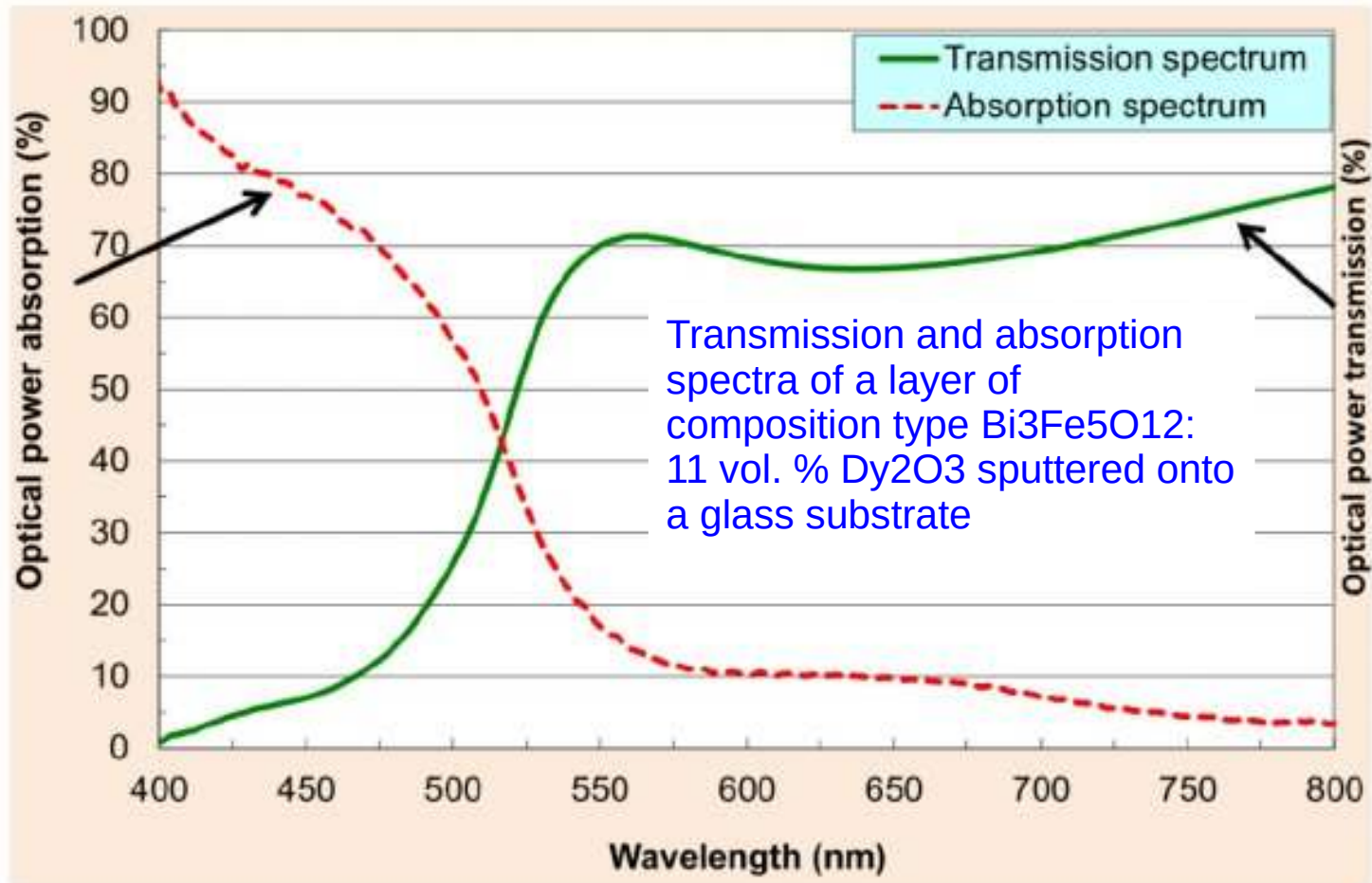
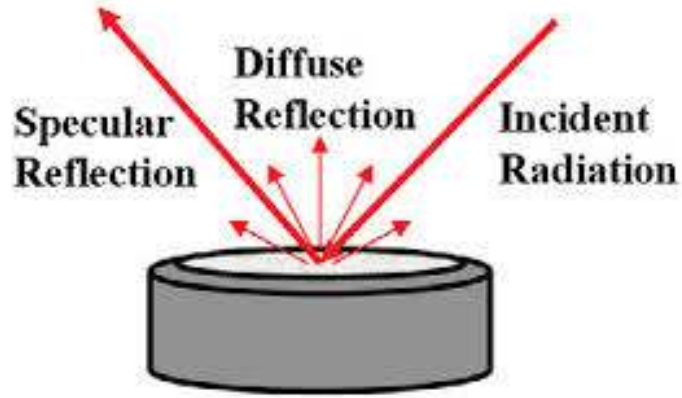


Figure 7: Shows a composite of UV-VIS range absorption spectra based on the comparative complexity of benzene and a bicinchoninic acid (BCA)-copper complex, and the effect it has on their respective absorption maxima (λ_{max}).

Abs. vs Trans.

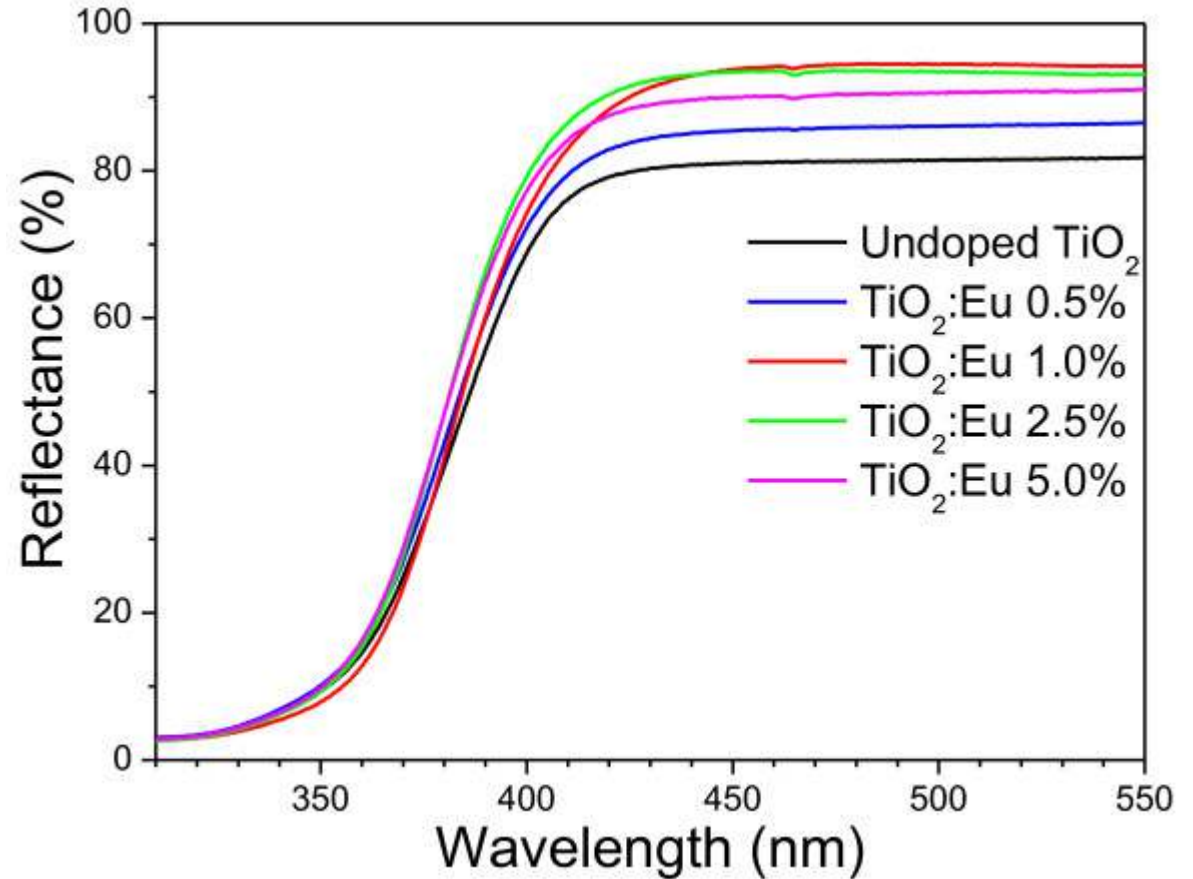


Diffuse reflectance mode

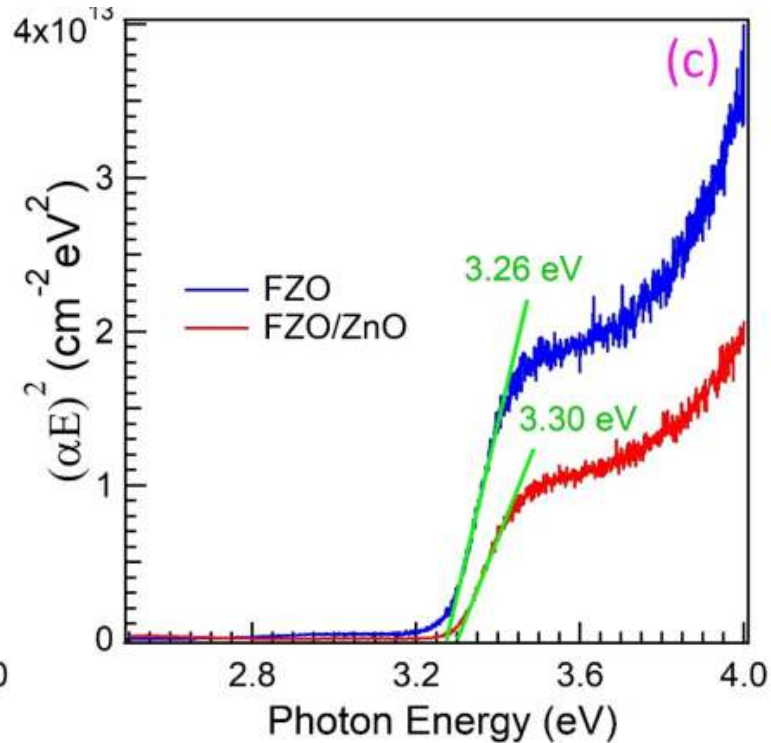
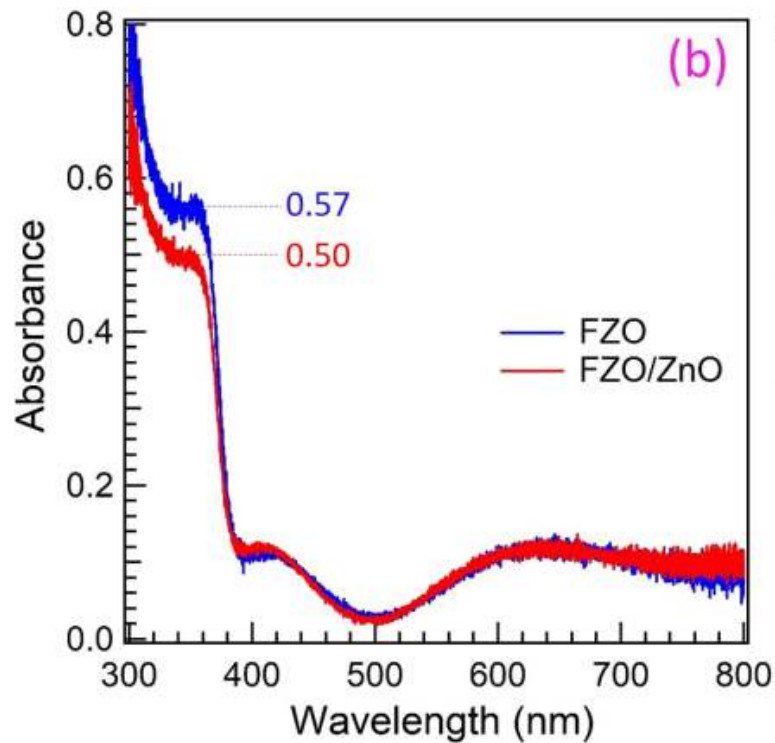


Kubelka-Munk transformation $F(R)$

$$F(R) = \alpha = \frac{(1 - R)^2}{2R}$$



Menentukan bandgap

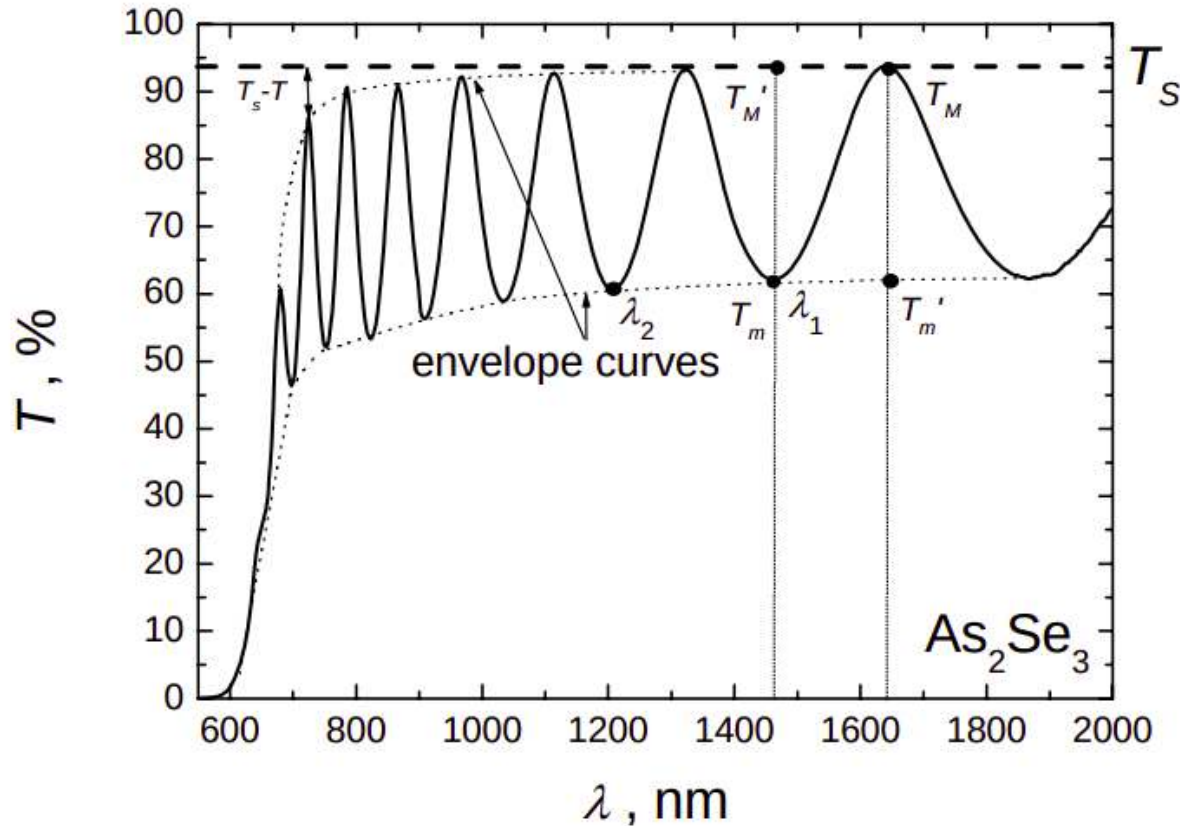


$$\alpha E = A(E - E_g)^n$$

n could be **1/2** , 3/2 , 2, and 3 for the allowed indirect, forbidden indirect, allowed direct, and forbidden direct transitions

Menentukan ketebalan

- Software: PARAV, PRISA



Ketebalan:

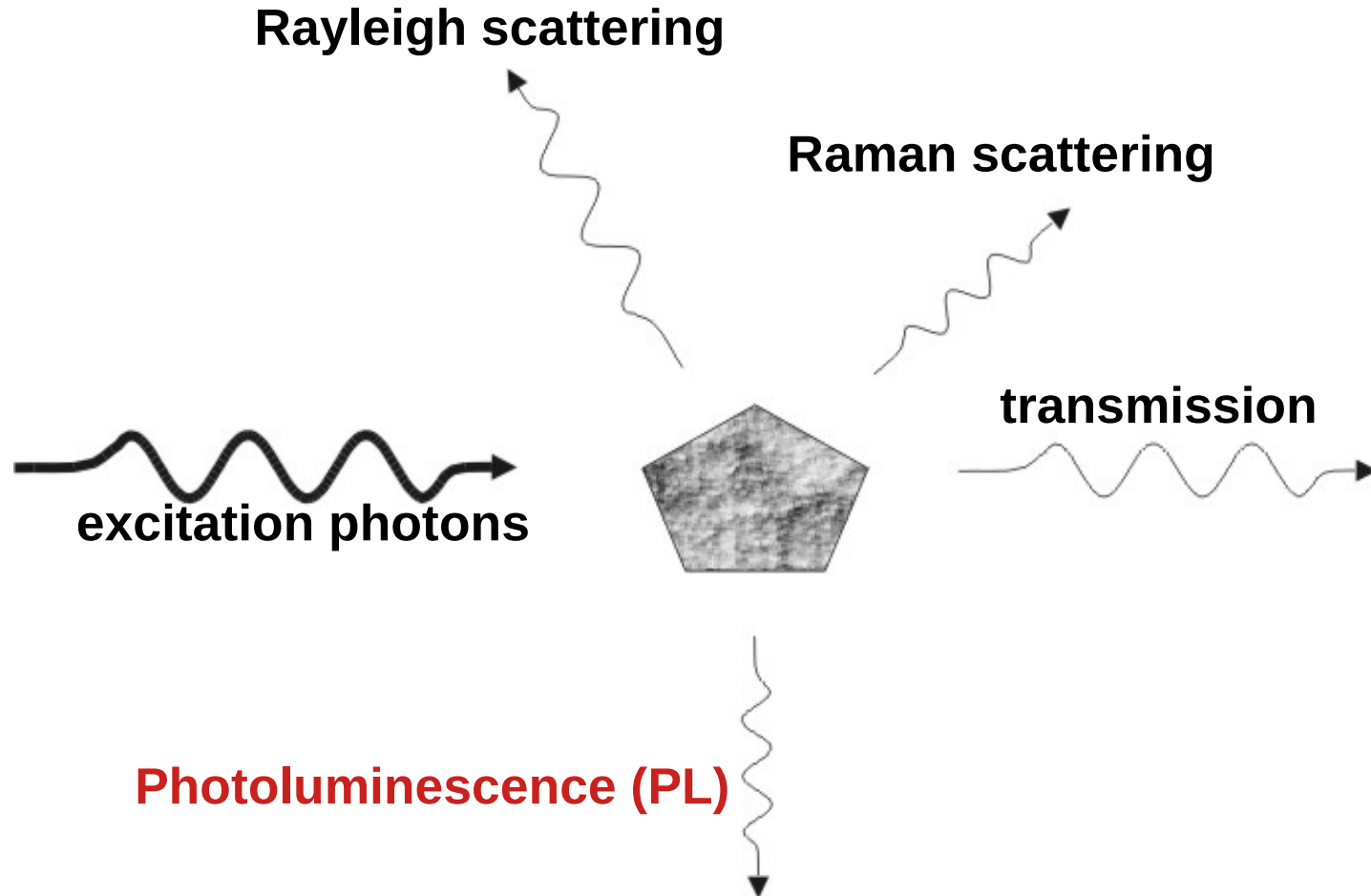
$$d = \frac{\lambda_1 \lambda_2}{2(\lambda_1 n_2 - \lambda_2 n_1)}$$

Indeks bias
(fully transmission region):

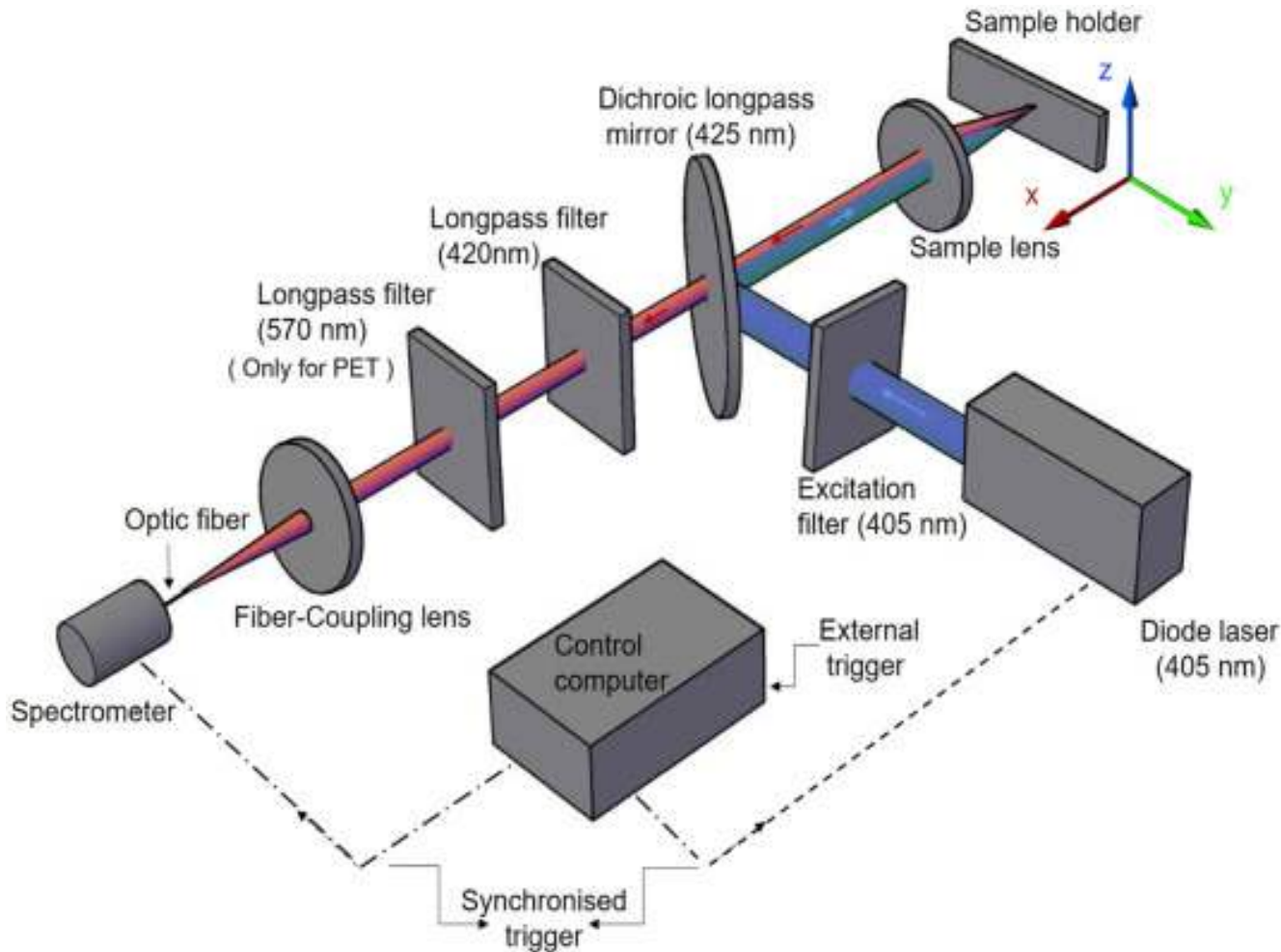
$$n = \left[M + (M^2 - n_s^2)^{1/2} \right]^{1/2}$$

$$M = \frac{2n_s}{T_m} - \frac{n_s^2 + 1}{2}$$

Interaksi Cahaya-Material



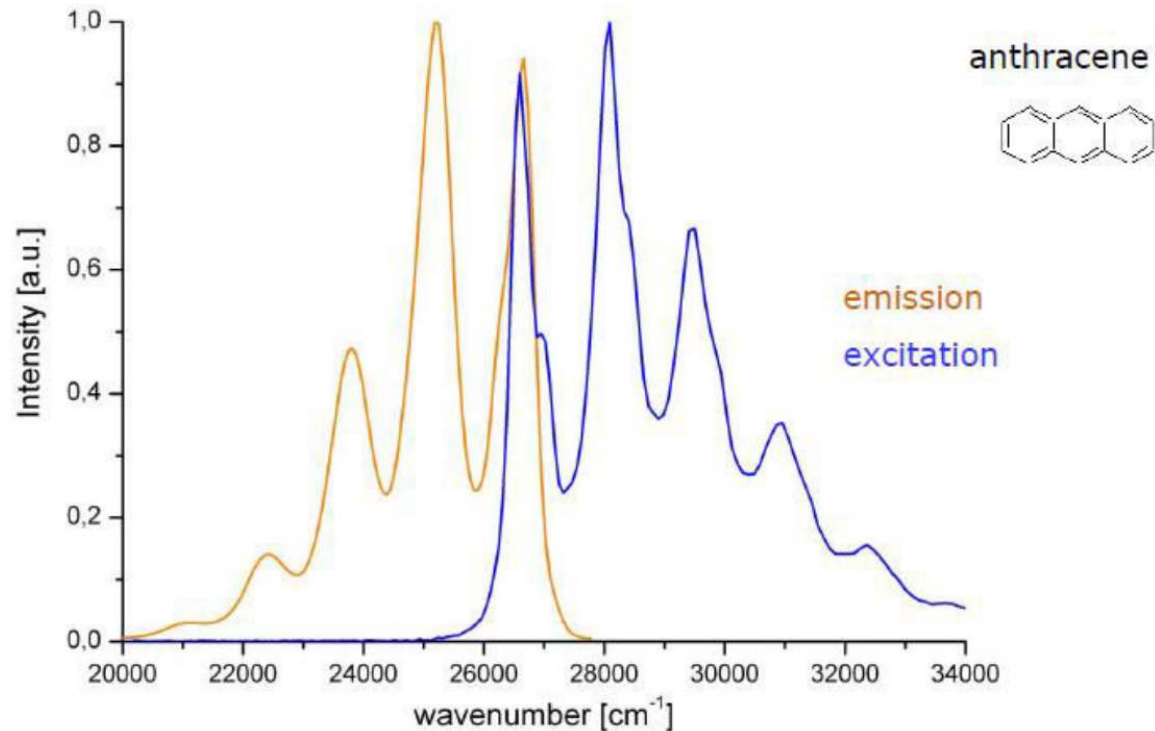
PL Spectroscopy



- excitation wavelength is fixed,
- emission intensity vs wavelength is obtained by scanning a monochromator of spectrometer.

The Stokes Shift

- The energy of emission is typically **less than** that of absorption.
- Fluorescence typically occurs at lower energies or longer wavelength.



Fluorescence life times and quantum yield

Quantum yield is the ratio of the number of photons emitted to the number absorbed.

$$Q = \frac{\Gamma}{\Gamma + k_{nr}}$$

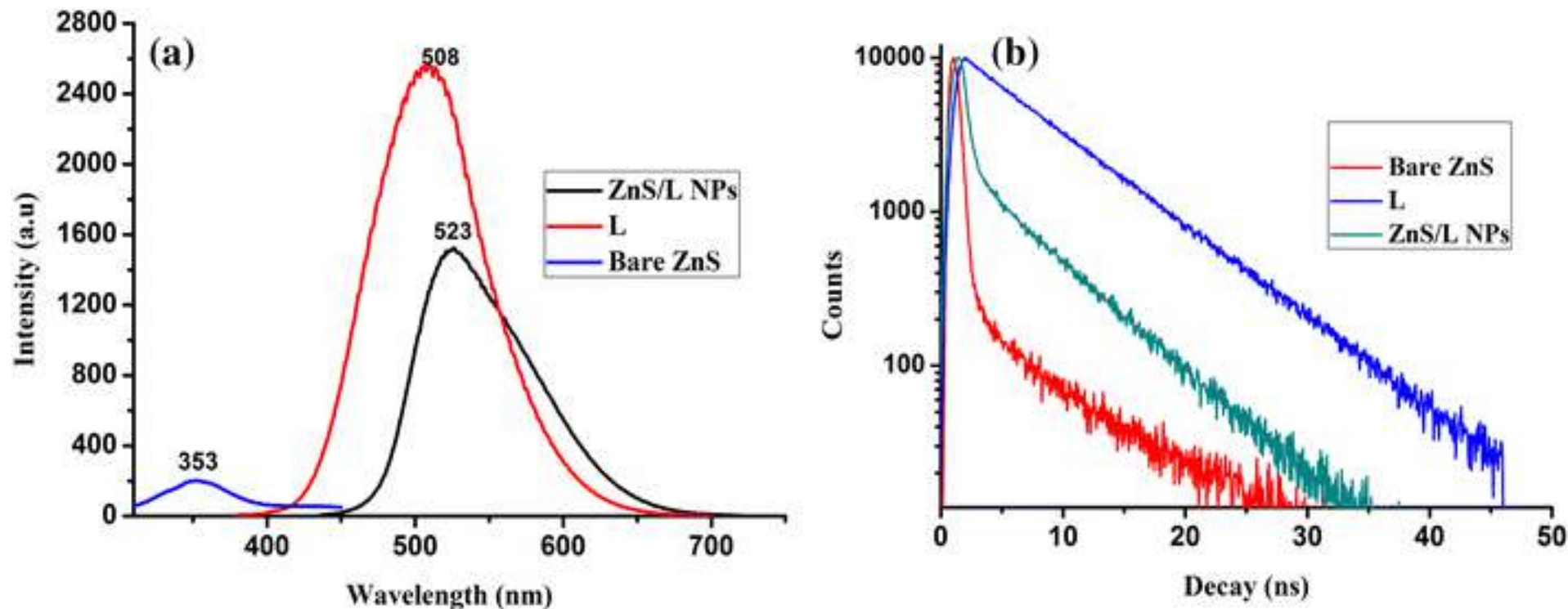
Γ = the emissive rate of fluorophore.

k_{nr} = rate of non-radiative decay

The lifetime of the excited state is defined by the **average time the molecule spends in the excited state** prior to the return to the ground state.

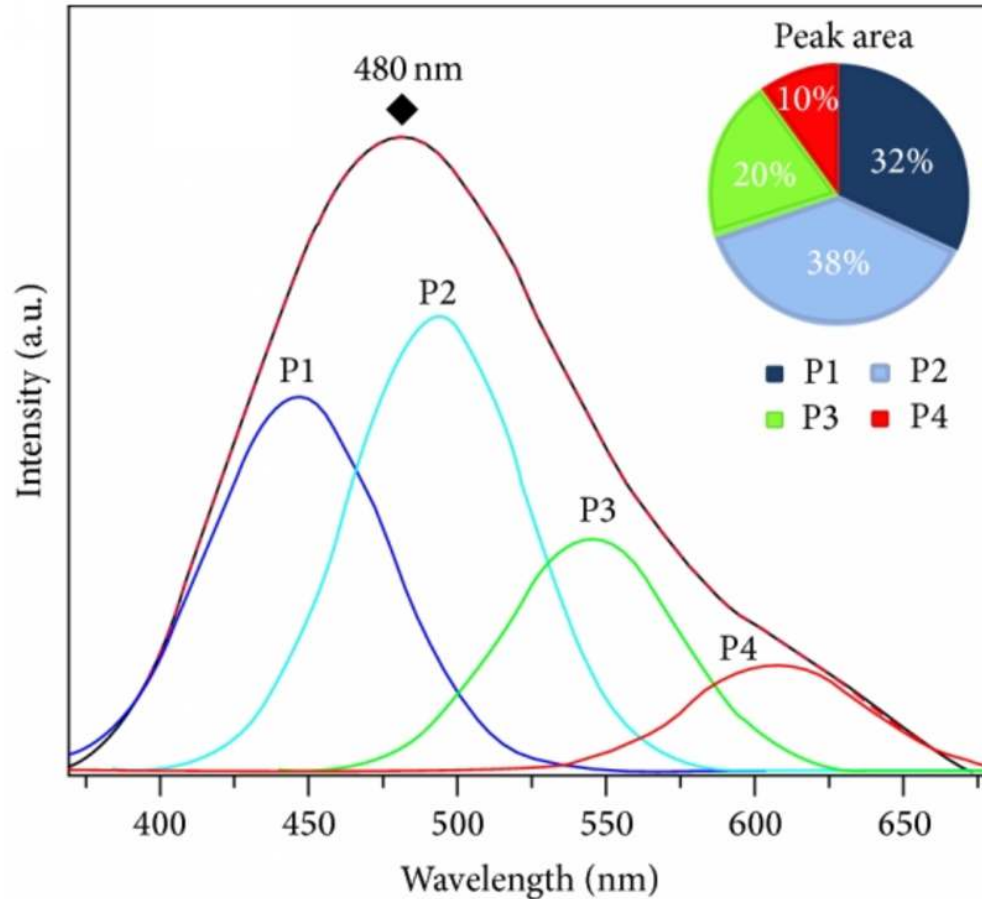
$$\tau = \frac{1}{\Gamma + k_{nr}}$$

Fluorescence lifetime



The PL spectra (left) and fluorescence lifetime (right) of bare ZnS, pure Ligand (L=2-[(2-hydroxy-ethyl)-(4-[2,20;60,200] terpyridin-40-yl-phenyl)-amino]-ethanol)) and the nanohybrids

Menentukan λ_{emisi}



Suatu emisi berisi dari berbagai panjang gelombang:
perlu dievaluasi